

Low health literacy and diabetes type 2: making a healthy lifestyle accessible to everyone

BSc Creative Technology thesis

Problem

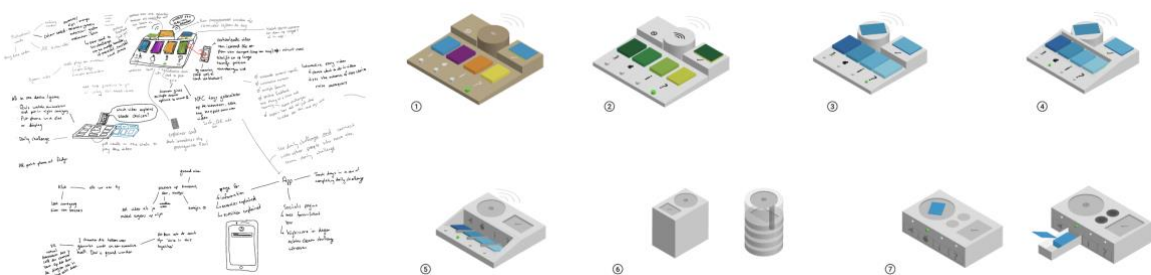
Many people with type 2 diabetes and low health literacy struggle to live healthier lives because they find health information hard to understand.

Research

I investigated strategies for effectively communicating with people with low health literacy. The main findings concerned simple language use, interactive communication, repeating information and using various means of communication.

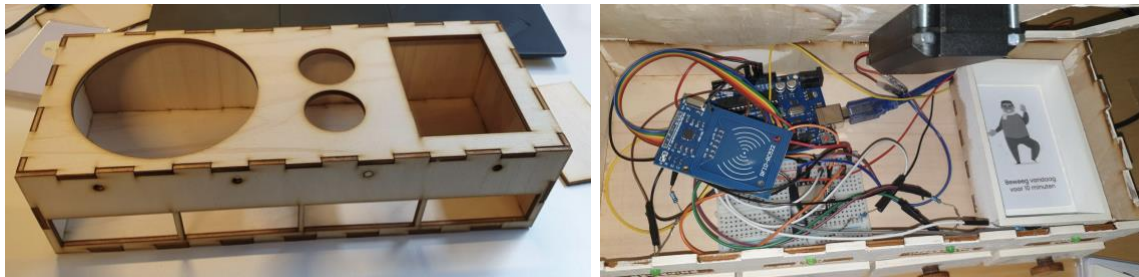
Concept & Design

I used the design thinking methodology with a diverging brainstorming and converging prototyping phase. This led to the creation of an interactive coaching box with multi-modal communication (visual, audio, and physical). Features included daily challenge cards with text and illustrations, audio playback, NFC-triggered animations, and physical interactivity.



Development

A physical prototype was built with a laser-cut casing and integrated sensors (LEDs, pressure sensor, NFC reader, speakers) controlled by Arduino.



Evaluation

I evaluated the prototype with visitors at a dietician's office. I specifically assessed the degree to which the box tailored to the audience, its learnability and aesthetic appeal.

Impact

The thesis showed that a simple, interactive, and multi-modal system can make health information more engaging and understandable for people with low health literacy.